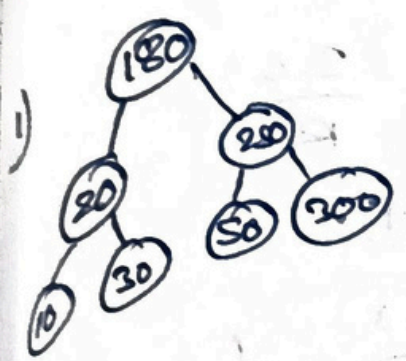


ms/26

CO-3 AT-3 Tree Traversal Challenge



* Preorder (NLR):

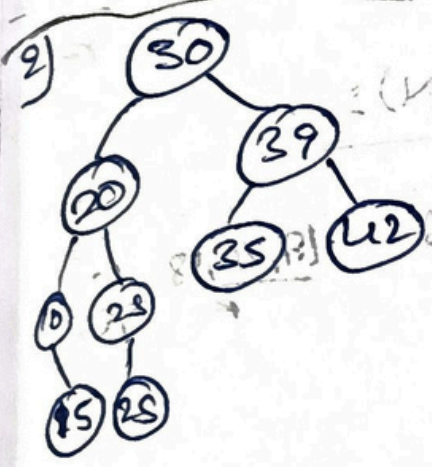
$\frac{180}{N}$ $\frac{20, 10, 30}{L}$ $\frac{250, 150, 200}{R}$

* Inorder (LNR):

$\frac{10, 20, 30}{L}$ $\frac{180}{N}$ $\frac{150, 250, 200}{R}$

* Postorder (LRN):

$\frac{10, 20, 30}{L}$ $\frac{150, 200, 250}{R}$ $\frac{180}{N}$



* Preorder (NLR):

$\frac{30}{N}$, $\frac{20, 10, 15, 25, 25}{L}$ $\frac{39, 35, 42}{R}$

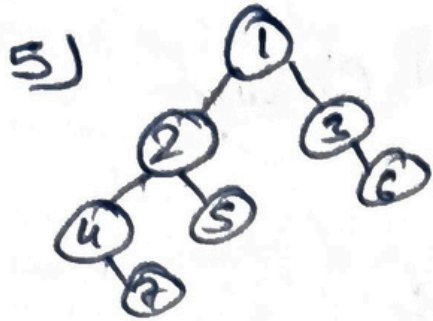
* Inorder (LNR):

$\frac{10, 15, 20, 25, 25}{L}$ $\frac{30}{N}$ $\frac{35, 39, 42}{R}$

* Postorder (LRN):

$\frac{15, 10, 25, 25, 20}{L}$ $\frac{20, 35, 42, 39}{R}$ $\frac{30}{N}$





* Preorder (NLR): $\frac{1}{N} \frac{2, 4, 7, 5}{L}$

* Inorder (LNR):

$\frac{7, 4, 2, 5}{L} \frac{1}{N} \frac{3, 6}{R}$

* Post order (LRN):

$\frac{7, 4, 5, 2}{L} \frac{6, 3}{R} \frac{1}{N}$

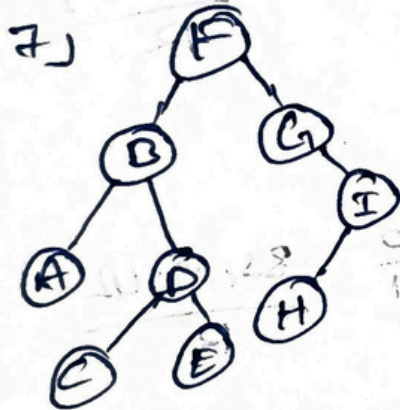


* Preorder (NLR):

$\frac{15}{N} \frac{6, 3, 2, 4, 7, 13, 9}{L} \frac{12, 20}{R}$

* Inorder (LRN):

$\frac{2, 4}{L} \frac{3, 9, 13, 7}{L} \frac{6, 15, 12, 20, 18}{R}$



* Inorder (LNR):

$\frac{A, B, C, D, E}{L} \frac{F}{N} \frac{G, H, I}{R}$

8) Given DBEFA GH C → Inorder

A B D E E C G H → Preorder

* Post order (LRN):

$\frac{D, F, E, B}{L} \frac{G, H, C}{R}$





* Inorder (LNR):

$$\frac{L, G, I}{L} \quad \frac{A}{N} \quad \frac{J, S, I}{R}$$

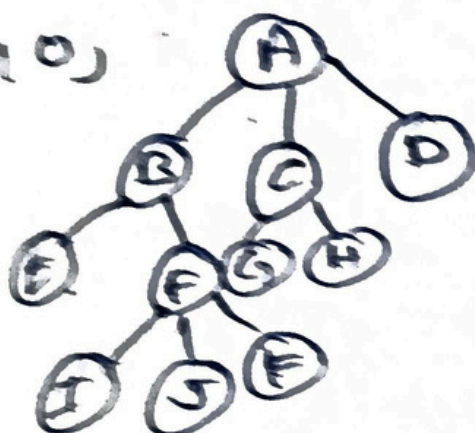
* Post order (RNL):

$$\frac{L, J, I}{L}, \frac{J, S, I}{R}, \frac{A}{N}$$

* Pre order (NLR):

$$\frac{A}{N} \quad \frac{G, I, J}{L} \quad \frac{S, I, I}{R}$$

10)



* Preorder:-

Root $\rightarrow$ children from L to R

A B E F I J K C G H D

A
 $\downarrow$
 B $\rightarrow$ E $\rightarrow$ F $\rightarrow$ I $\rightarrow$ J $\rightarrow$ K
 $\downarrow$
 C $\rightarrow$ G $\rightarrow$ H
 $\downarrow$
 D

* Post order (LRN):

$$\frac{E \ I \ J \ K \ F \ B \ G \ H \ C \ D \ A}{N}$$

~~* Inorder:-~~

Not applicable (Only in binary tree)